

PROBLEM SHEET 3: SOLUTIONS

1. (a) For small u , the interaction term behaves like auv , i.e. like standard Lotka–Volterra predator–prey. For large u , the interaction term behaves like av , i.e. like prey consumption rate is only dependent on the predator population, rather than the opportunities for consumption between predator and prey. Just like in the spruce budworm model, this is saying that the *predators get full*.

- (b) Looking for equilibria, we say

$$0 = u \left(1 - \frac{u}{k} \right) - \frac{auv}{1+u}, \quad (1a)$$

$$0 = -bv + \frac{auv}{1+u}. \quad (1b)$$

Looking first at eq. (1b), we have

$$0 = v \left(-b + \frac{au}{1+u} \right) \implies v = 0, \quad b = \frac{au}{1+u}. \quad (2)$$

Plugging the first solution, $v = 0$, into eq. (1a), we get $u = 0$ or $u = k$, so we have found $(0, 0)$ and $(k, 0)$.

The second solution rearranges to

$$u = \frac{b}{a-b}. \quad (3)$$

Rearranging eq. (1a) for v and substituting this solution for u in, we get

$$v = \frac{1+u}{a} \left[1 - \frac{u}{k} \right] = \dots = \frac{k(a-b) - b}{k(a-b)^2}, \quad (4)$$

and so our final equilibrium is

$$(u^*, v^*) = \left(\frac{b}{a-b}, \frac{k(a-b) - b}{k(a-b)^2} \right). \quad (5)$$

This is permissible if $a > b$ and $k(a-b) \geq b$.

- (c) If $a = 2$ and $b = 1$ then

$$(u^*, v^*) = \left(1, \frac{k-1}{k} \right), \quad (6)$$

and this is permissible if $k \geq 1$.

- (d) Under this choice for a and b , the Jacobian is

$$\mathbf{J} = \begin{pmatrix} 1 - \frac{2u}{k} - \frac{2v}{(1+u)^2} & -\frac{2u}{1+u} \\ \frac{2v}{(1+u)^2} & -1 + \frac{2u}{1+u} \end{pmatrix}. \quad (7)$$

Evaluated at each equilibrium, we have:

- at $(0, 0)$:

$$\mathbf{J}(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \implies \lambda = 1, -1. \quad (8)$$

So this is unstable (a saddle point).

- at $(k, 0)$:

$$\mathbf{J}(k, 0) = \begin{pmatrix} -1 & -\frac{k}{1+k} \\ 0 & -1 + \frac{2k}{1+k} \end{pmatrix} \implies \lambda = -1, -1 + \frac{2k}{1+k}. \quad (9)$$

This is unstable if the second eigenvalue is positive, i.e. if

$$-1 + \frac{2k}{1+k} > 0 \implies k > 1. \quad (10)$$

So it is stable for $0 < k < 1$ (recall k is a positive constant), and unstable for $k > 1$.

- at $(u^*, v^*) = (1, (k-1)/k)$:

$$\mathbf{J}(u^*, v^*) = \begin{pmatrix} 1 - \frac{2}{k} - \frac{1}{2} \left(\frac{k-1}{k} \right) & -1 \\ \frac{1}{2} \left(\frac{k-1}{k} \right) & 0 \end{pmatrix} = \begin{pmatrix} \frac{k-3}{2k} & -1 \\ \frac{k-1}{2k} & 0 \end{pmatrix}. \quad (11)$$

Following the hint, we have

$$\det \mathbf{J} = \frac{k-1}{2k}, \quad \text{tr } \mathbf{J} = \frac{k-3}{2k}, \quad (\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = \frac{-7k^2 + 2k + 9}{4k^2}. \quad (12)$$

Now we can see the determinant and trace change signs at $k = 1$ and $k = 3$, so let's think about the regions either side and between these points:

- If $0 < k < 1$: the equilibrium is not permissible.
- If $1 < k < 3$: $\det \mathbf{J} > 0$, $\text{tr } \mathbf{J} < 0$, so the equilibrium is stable.
- If $k > 3$: $\det \mathbf{J} > 0$, $\text{tr } \mathbf{J} > 0$, so the equilibrium is unstable.

We can put this information all together:

u_0	v_0	permissible?	stable?	unstable?
0	0	always	—	always
k	0	always	$0 < k < 1$	$k > 1$
1	$\frac{k-1}{k}$	$k > 1$	$1 < k < 3$	$k > 3$

- (e) Look again at the table above. Ignoring the complete extinction equilibrium, we have

$$(u_0, v_0) = (k, 0) \quad \text{and} \quad \left(1, \frac{k-1}{k} \right). \quad (13)$$

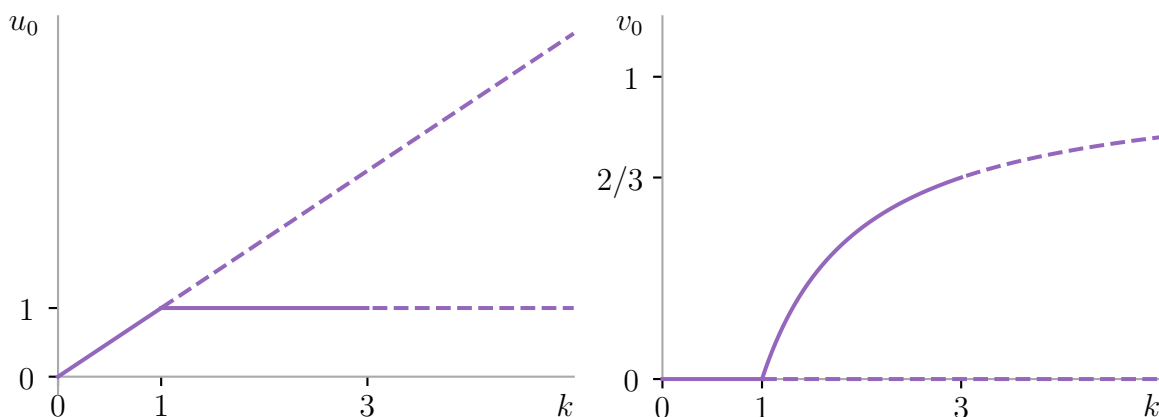
So for the u_0 bifurcation diagram we have:

- $u_0 = k$, permissible for all k and stable for $0 < k < 1$, unstable otherwise,
- $u_0 = 1$, permissible only for $k > 1$, stable for $1 < k < 3$, unstable otherwise.

And for the v_0 bifurcation diagram we have:

- $v_0 = 0$, permissible for all k and stable for $0 < k < 1$, unstable otherwise,
- $v_0 = (k - 1)/k$, permissible only for $k > 1$, stable for $1 < k < 3$, unstable otherwise.

These look on the graph like:

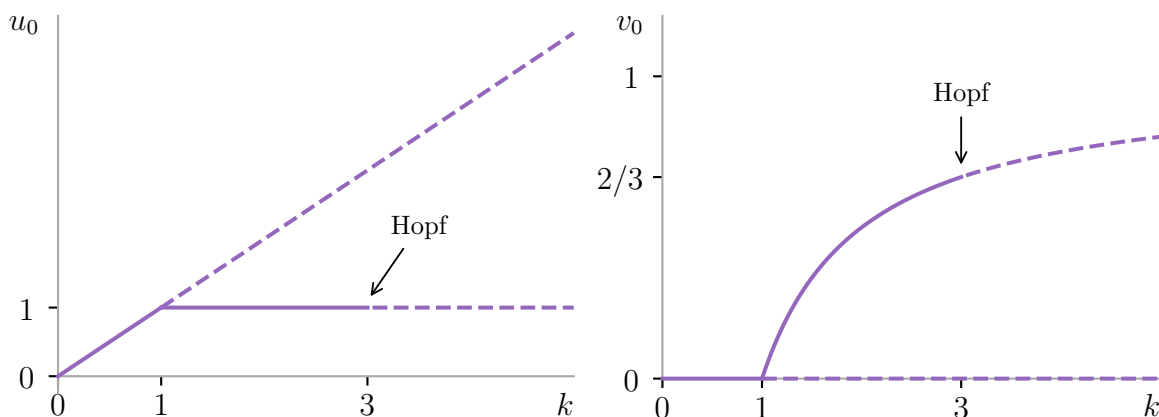


- (f) The Hopf bifurcation is the point where a stable spiral becomes an unstable spiral (or the other way round).

Looking at the graph, there are two bifurcations where stability changes: one at $k = 1$ and $k = 3$.

At $k = 1$, $(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = 1 > 0$, so the bifurcation there is a transition between nodes, not spirals (look at the tr/det table in the notes).

At $k = 3$, $(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = -4/3 < 0$, so this is the bifurcation between spirals. Hence this is the Hopf bifurcation and we can mark it on our diagrams:



2. (a) We start with the advection–diffusion equation, which in one dimension is

$$\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}[uv] - \frac{\partial J}{\partial x} + f \quad (14)$$

where J is the flux, f a source and v the bulk velocity.

- The question states $v = 0$.
- Fick’s law in one dimension is $J = -D\partial u/\partial x$, with D the diffusion constant.
- Finally, the bacteria death mentioned in the text is clearly a source term $-\lambda u$ (or a sink as it is sometimes called when negative).

Put together, this gives the required equation.

The boundary conditions will be at $x = 0$ and $x = L$. We put the ‘death’ condition at $x = 0$ which requires the population is zero at this point, $u(0, t) = 0$. The density gradient condition requires that $\partial u/\partial x(L, t) = \mu$.

This is the correct sign on μ because Fick’s law states that density flows from higher to lower density. At $x = L$, we want the bacteria to come in (move to the left), so we want more concentration on the right, i.e. a positive gradient.

- (b) The equilibrium requires that

$$D\frac{d^2u}{dx^2} - u = 0. \quad (15)$$

We either write down the auxiliary equation in m , or just go straight for exponential solutions, $u = e^{mx}$, giving us

$$Dm^2 - 1 = 0 \implies m = \pm \frac{1}{\sqrt{D}}. \quad (16)$$

So the general solution is

$$u(x) = \tilde{A}e^{-x/\sqrt{D}} + \tilde{B}e^{x/\sqrt{D}}, \quad (17)$$

although the solution in the question suggests we should write it in terms of hyperbolic trig functions,

$$u(x) = A \cosh\left(\frac{x}{\sqrt{D}}\right) + B \sinh\left(\frac{x}{\sqrt{D}}\right). \quad (18)$$

The first boundary condition at $x = 0$ requires $A = 0$ so

$$u(x) = B \sinh\left(\frac{x}{\sqrt{D}}\right). \quad (19)$$

The x -derivative is

$$\frac{du}{dx} = \frac{B}{\sqrt{D}} \cosh\left(\frac{x}{\sqrt{D}}\right), \quad (20)$$

and the second boundary condition reads

$$\frac{B}{\sqrt{D}} \cosh\left(\frac{L}{\sqrt{D}}\right) = \mu \implies B = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})}. \quad (21)$$

So the final solution is

$$u(x) = \frac{\sqrt{D}\mu \sinh(x/\sqrt{D})}{\cosh(L/\sqrt{D})}, \quad (22)$$

i.e.

$$\alpha = \sqrt{D}, \quad \beta = \frac{L}{\sqrt{D}}. \quad (23)$$

(c) To find the centre of mass we must integrate,

$$\int_0^L xu(x) dx = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})} \int_0^L x \sinh\left(\frac{x}{\sqrt{D}}\right) dx. \quad (24)$$

We use integration by parts to get

$$\int_0^L x \sinh\left(\frac{x}{\sqrt{D}}\right) dx = \sqrt{D} \left[x \cosh\left(\frac{x}{\sqrt{D}}\right) \right]_0^L - \sqrt{D} \int_0^L \cosh\left(\frac{x}{\sqrt{D}}\right) dx \quad (25)$$

$$= \sqrt{D}L \cosh\left(\frac{L}{\sqrt{D}}\right) - D \left[\sinh\left(\frac{x}{\sqrt{D}}\right) \right]_0^L \quad (26)$$

$$= \sqrt{D}L \cosh\left(\frac{L}{\sqrt{D}}\right) - D \sinh\left(\frac{L}{\sqrt{D}}\right). \quad (27)$$

The centre of mass, C_m , is given by

$$C_m = \frac{\int_0^L xu(x) dx}{\int_0^L u(x) dx}, \quad (28)$$

and

$$\int_0^L u(x) dx = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})} \int_0^L \sinh\left(\frac{x}{\sqrt{D}}\right) dx \quad (29)$$

$$= \frac{D\mu}{\cosh(L/\sqrt{D})} \left[\cosh\left(\frac{L}{\sqrt{D}}\right) - 1 \right]. \quad (30)$$

So

$$C_m = \frac{L \cosh(L/\sqrt{D}) - \sqrt{D} \sinh(L/\sqrt{D})}{\cosh(L/\sqrt{D}) - 1}, \quad (31)$$

i.e.

$$\alpha = \sqrt{D}, \quad \beta = \frac{L}{\sqrt{D}}, \quad \gamma = L, \quad \delta = 1. \quad (32)$$

(d) In this limit $L/\sqrt{D} \gg 1$. Hence $\sinh(L/\sqrt{D}) \approx \cosh(L/\sqrt{D})$ and $\cosh(L/\sqrt{D}) \gg 1$ so $C_m \approx L$ and the whole distribution gets squashed against the boundary at $x = L$.